

COMPLETE STUDY GUIDE

Mastering *Trigonometry*

From absolute zero to calculus-ready — explained in plain
English with no shortcuts.

01 What Is Trig?

02 Angles & Radians

03 SOH-CAH-TOA

04 Unit Circle

05 Special Angles

06 Graphs

07 Identities

08 More Formulas

09 Inverse Trig

10 Calculus Prep

1

BASIC

What Is Trigonometry?

The study of triangles — and secretly, the study of waves

Trigonometry means "triangle measurement" (Greek: **trigonon** = triangle, **metron** = measure).
It answers a simple question:

 KEY IDEA

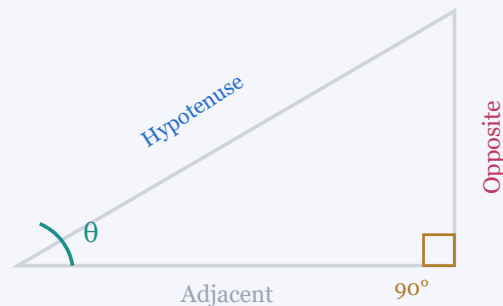
If you know one angle and one side of a right triangle, can you find all the other sides?

Yes — trigonometry is exactly how.

Those same relationships also describe **waves, circles, oscillations, sound, light, and electricity** — making trig the backbone of physics, engineering, and calculus.

The Right Triangle

- **One 90° angle** (marked with a small square)
- **Two acute angles** (both $< 90^\circ$, always sum to 90°)
- **Three sides** named relative to your chosen angle θ



Right triangle with angle θ at bottom-left

2

BASIC

Angles: Degrees & Radians

Two ways to measure — why radians rule calculus

Degrees: full rotation = **360°** . Radians: full rotation = **2π** . Since circumference of unit circle = 2π , a full rotation = 2π radians.

MASTER CONVERSION

$$180^\circ = \pi \text{ radians}$$

Degrees $\rightarrow$ Radians: multiply by $\pi/180$

Radians $\rightarrow$ Degrees: multiply by $180/\pi$

DEGREES	RADIANS	DEGREES	RADIANS
0°	0	90°	$\pi/2$
30°	$\pi/6$	180°	π
45°	$\pi/4$	270°	$3\pi/2$
60°	$\pi/3$	360°	2π

└ CALCULUS CONNECTION

Calculus is always in radians. Every derivative/integral formula for trig only works in radians. The reason: $\lim_{x \rightarrow 0} \sin(x)/x = 1$ only holds in radians.

3

BASIC

SOH-CAH-TOA — The Big Three

Sine, Cosine, Tangent defined simply

SOH-CAH-TOA

SOH: $\sin(\theta) = \text{Opposite} / \text{Hypotenuse}$

CAH: $\cos(\theta) = \text{Adjacent} / \text{Hypotenuse}$

TOA: $\tan(\theta) = \text{Opposite} / \text{Adjacent}$

SINE

$$\sin \theta = \text{opp/hyp}$$

"How high up"

COSINE

$$\cos \theta = \text{adj/hyp}$$

"How far across"

TANGENT

$$\tan \theta = \text{opp/adj}$$

"The slope"

Reciprocal Functions

COSECANT

$$\csc \theta = 1/\sin \theta$$

hyp/opp

SECANT

$$\sec \theta = 1/\cos \theta$$

hyp/adj

COTANGENT

$$\cot \theta = 1/\tan \theta$$

adj/opp



TIP

$\tan(x) = \sin(x)/\cos(x)$. Tangent is undefined when $\cos(x) = 0$, i.e., at $x = \pm\pi/2, \pm3\pi/2, \dots$

4

BASIC

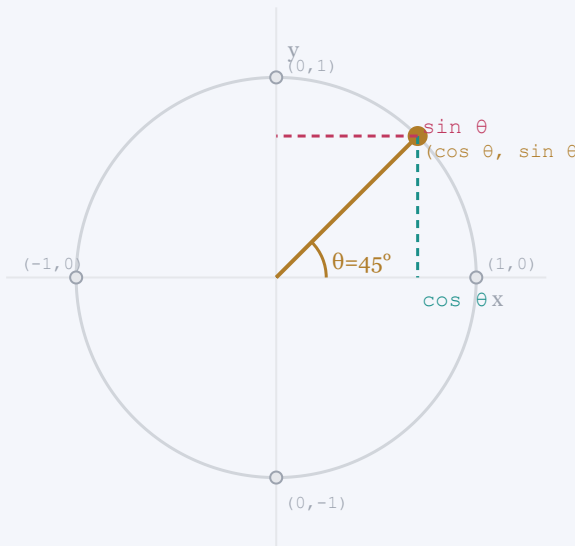
The Unit Circle

The single most important picture in trigonometry



KEY IDEA

For any angle θ on the unit circle, the point where it lands = **$(\cos \theta, \sin \theta)$** .



Every point on the unit circle = $(\cos \theta, \sin \theta)$

QUADRANT	RANGE	SIN	COS	TAN
I (top right)	$0 \rightarrow \pi/2$	+	+	+
II (top left)	$\pi/2 \rightarrow \pi$	+	-	-
III (bottom left)	$\pi \rightarrow 3\pi/2$	-	-	+
IV (bottom right)	$3\pi/2 \rightarrow 2\pi$	-	+	-



TIP

"All Students Take Calculus" — counterclockwise from Q1: All positive → Sine → Tangent → Cosine positive.

5

INTERMEDIATE

Special Angles

Values every calculus student must know cold

45-45-90 TRIANGLE

Sides: 1, 1, $\sqrt{2}$
(half of a square)

30-60-90 TRIANGLE

Sides: 1, $\sqrt{3}$, 2
(half of equilateral)

ANGLE (°)	RADIANS	SIN	COS	TAN
0°	0	0	1	0
30°	$\pi/6$	1/2	$\sqrt{3}/2$	$\sqrt{3}/3$
45°	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\pi/3$	$\sqrt{3}/2$	1/2	$\sqrt{3}$
90°	$\pi/2$	1	0	undefined



TIP

Pattern for sin: $\sqrt{0}/2, \sqrt{1}/2, \sqrt{2}/2, \sqrt{3}/2, \sqrt{4}/2 = 0, 1/2, \sqrt{2}/2, \sqrt{3}/2, 1$. Cosine is the same list reversed.

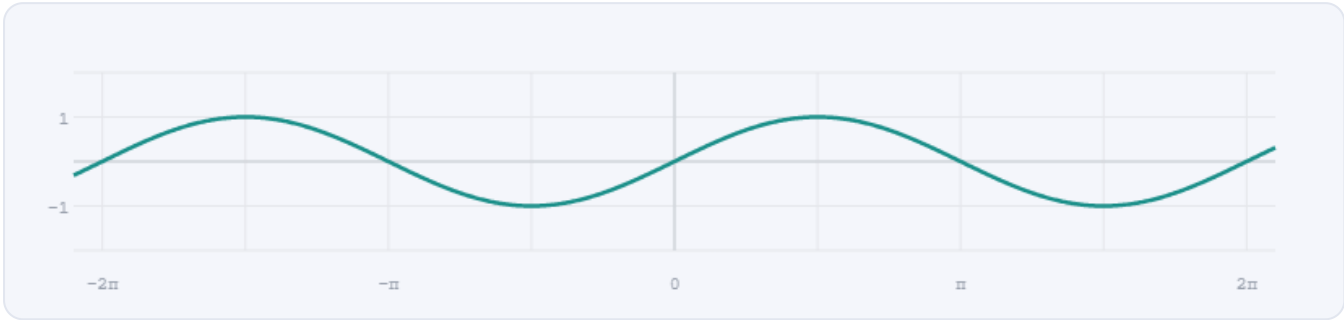
6

INTERMEDIATE

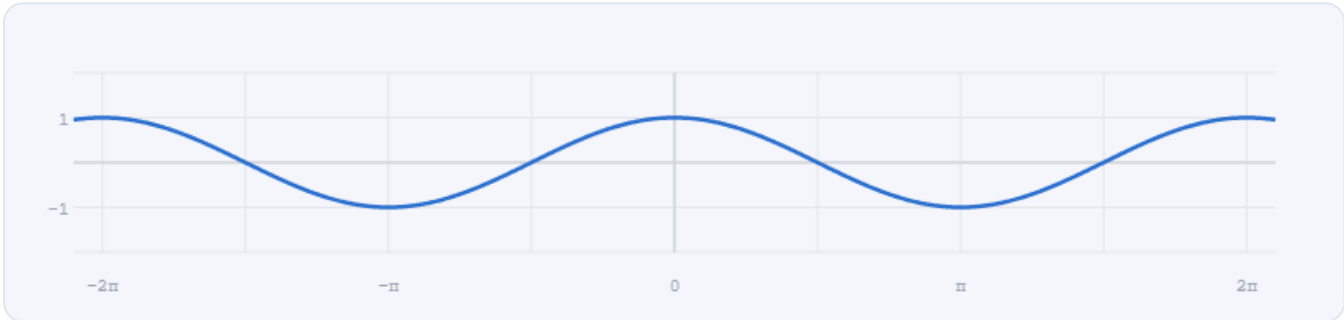
Graphs of Trig Functions

Visualizing waves — the language of calculus

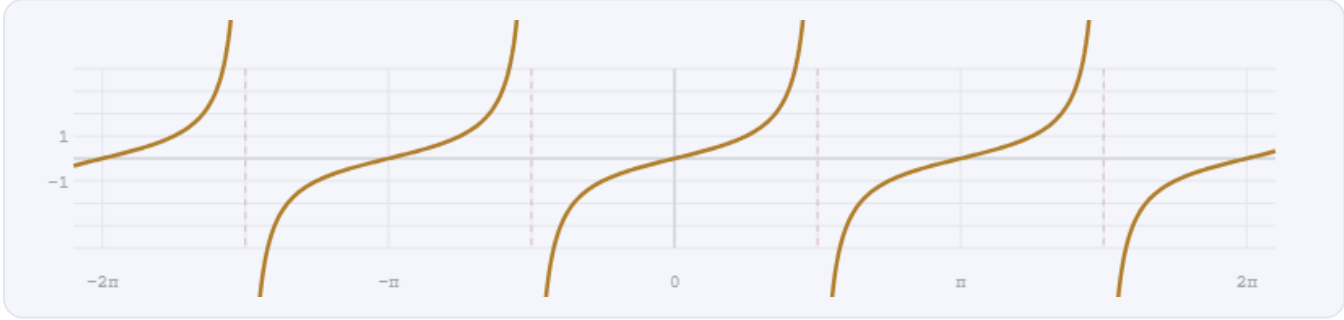
Sine Wave



Cosine Wave



Tangent Function



FUNCTION	PERIOD	AMPLITUDE	RANGE
$\sin(x)$	2π	1	$[-1, 1]$
$\cos(x)$	2π	1	$[-1, 1]$
$\tan(x)$	π	∞	$(-\infty, +\infty)$

GENERAL FORM

$$f(x) = A \cdot \sin(Bx + C) + D$$

A = Amplitude | Period = $2\pi/|B|$ | Phase Shift = $-C/B$ | D = Vertical Shift

 TIP

Even/Odd: $\cos(-x) = \cos(x)$ [even, y-axis symmetry]. $\sin(-x) = -\sin(x)$ [odd, origin symmetry].

7

INTERMEDIATE

Trigonometric Identities

The toolkit that makes every trig calculation possible

PYTHAGOREAN IDENTITIES

$$\sin^2(x) + \cos^2(x) = 1 \leftarrow \text{THE most important identity}$$

$$1 + \tan^2(x) = \sec^2(x) \leftarrow \text{divide above by } \cos^2(x)$$

$$1 + \cot^2(x) = \csc^2(x) \leftarrow \text{divide above by } \sin^2(x)$$

 KEY IDEA

$\sin^2(x) + \cos^2(x) = 1$. Comes from $x^2 + y^2 = 1$ on the unit circle. Every other identity derives from this.

QUOTIENT & RECIPROCAL

$$\tan(x) = \sin(x)/\cos(x) \quad \cot(x) = \cos(x)/\sin(x)$$

$$\csc(x) = 1/\sin(x) \quad \sec(x) = 1/\cos(x)$$

NEGATIVE ANGLE (EVEN/ODD)

$$\sin(-x) = -\sin(x) \quad \cos(-x) = \cos(x) \quad \tan(-x) = -\tan(x)$$

CO-FUNCTION IDENTITIES

$$\sin(\pi/2 - x) = \cos(x) \quad \cos(\pi/2 - x) = \sin(x)$$

$$\tan(\pi/2 - x) = \cot(x) \quad \sec(\pi/2 - x) = \csc(x)$$

CALCULUS CONNECTION

In calculus: rewrite $\sin^2(x)$ as $(1 - \cos^2 x)$ or $\cos^2(x)$ as $(1 - \sin^2 x)$ to crack integrals wide open.

8

ADVANCED

Sum, Difference & Double-Angle

Powerful tools for expanding and simplifying

SUM & DIFFERENCE

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = (\tan A \pm \tan B) / (1 \mp \tan A \cdot \tan B)$$

DOUBLE ANGLE (SET B = A ABOVE)

$$\sin(2A) = 2 \sin A \cos A$$

$$\cos(2A) = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$$

$$\tan(2A) = 2 \tan A / (1 - \tan^2 A)$$

POWER-REDUCING — CRITICAL FOR CALCULUS

$$\sin^2(x) = (1 - \cos 2x) / 2$$

$$\cos^2(x) = (1 + \cos 2x) / 2$$

CALCULUS CONNECTION

$\int \sin^2(x) dx$ = $\int (1/2 - 1/2 \cos 2x) dx = 1/2 x - 1/4 \sin(2x) + C$. Power-reducing makes this possible.

HALF-ANGLE

$$\sin(A/2) = \pm \sqrt{(1 - \cos A)/2}$$

$$\cos(A/2) = \pm \sqrt{(1 + \cos A)/2}$$

$$\tan(A/2) = \sin A / (1 + \cos A) = (1 - \cos A) / \sin A$$

9

ADVANCED

Inverse Trig Functions

Going backwards — from ratio to angle

⚠ WATCH OUT

$\sin^{-1}(x)$ means the inverse function, NOT $1/\sin(x)$. To write the reciprocal, write $\csc(x)$.

FUNCTION	NOTATION	DOMAIN	RANGE
Inverse Sine	$\arcsin(x)$	$[-1, 1]$	$[-\pi/2, \pi/2]$
Inverse Cosine	$\arccos(x)$	$[-1, 1]$	$[0, \pi]$
Inverse Tangent	$\arctan(x)$	$(-\infty, +\infty)$	$(-\pi/2, \pi/2)$

DERIVATIVES OF INVERSE TRIG

$$d/dx [\arcsin(x)] = 1 / \sqrt{1 - x^2}$$

$$d/dx [\arccos(x)] = -1 / \sqrt{1 - x^2}$$

$$d/dx [\arctan(x)] = 1 / (1 + x^2)$$

$$d/dx [\operatorname{arcsec}(x)] = 1 / (|x| \sqrt{x^2 - 1})$$

$\int$ CALCULUS CONNECTION

$\int 1/(1+x^2) dx = \arctan(x) + C$. $\int 1/\sqrt{1-x^2} dx = \arcsin(x) + C$. Recognize these patterns!

10

CALCULUS

Trig in Calculus

Where everything comes together

FUNDAMENTAL TRIG LIMITS

$$\lim_{x \rightarrow 0} \sin(x)/x = 1 \leftarrow \text{Proves } d/dx[\sin x] = \cos x$$

$$\lim_{x \rightarrow 0} (1 - \cos x)/x = 0$$

KEY IDEA

These limits only equal 1 and 0 in radians. This is the deepest reason calculus uses radians.

SIX TRIG DERIVATIVES

$$d/dx [\sin x] = \cos x \quad d/dx [\cos x] = -\sin x$$

$$d/dx [\tan x] = \sec^2 x \quad d/dx [\cot x] = -\csc^2 x$$

$$d/dx [\sec x] = \sec x \cdot \tan x \quad d/dx [\csc x] = -\csc x \cdot \cot x$$

Memory trick: the three "co-" functions (cosine, cotangent, cosecant) all carry a **negative sign** in their derivatives.

BASIC TRIG INTEGRALS

$$\int \sin x \, dx = -\cos x + C \quad \int \cos x \, dx = \sin x + C$$

$$\int \tan x \, dx = \ln|\sec x| + C \quad \int \cot x \, dx = \ln|\sin x| + C$$

$$\int \sec x \, dx = \ln|\sec x + \tan x| + C$$

$$\int \csc x \, dx = -\ln|\csc x + \cot x| + C$$

Trig Substitution for Integrals

EXPRESSION	USE	IDENTITY
$\sqrt{a^2 - x^2}$	$x = a \cdot \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \cdot \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$

EXPRESSION	USE	IDENTITY
$\sqrt{x^2 - a^2}$	$x = a \cdot \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$

TAYLOR SERIES

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\text{Euler's Formula: } e^{i\theta} = \cos \theta + i \cdot \sin \theta$$

✓ Calculus Trig Checklist

- Convert freely between degrees and radians
- State $\sin$, $\cos$, $\tan$ for 0 , $\pi/6$, $\pi/4$, $\pi/3$, $\pi/2$ from memory
- Sketch $\sin$, $\cos$, $\tan$ and describe all key properties
- State $\sin^2 x + \cos^2 x = 1$ and both derived Pythagorean identities
- Apply $\sin(A \pm B)$ and $\cos(A \pm B)$ formulas
- Write $\sin(2x)$ and $\cos(2x)$ in multiple equivalent forms
- Rewrite $\sin^2 x$ and $\cos^2 x$ using power-reducing formulas
- Know domains and ranges of $\arcsin$, $\arccos$, $\arctan$
- Evaluate $\lim_{x \rightarrow 0} \sin(x)/x = 1$ and explain why
- State all 6 trig derivatives from memory
- Choose the right trig substitution for each radical form